Kartik Narayan

LinkedIn | Github | Google Scholar
https://kartik-3004.github.io/portfolio/

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Summary

I am a 3rd year (final year) Ph.D. student in the Computer Science Department at Johns Hopkins University. My current research interests lies in multimodal LLMs, agents, long-horizon tool calling, agentic post-training, agentic search, reasoning and video generation.
Graduation Date: December 2026

Education

Johns Hopkins University — Baltimore, MD
Ph.D. Computer Science (GPA: 4.0/4.0), Advisor: Dr.Vishal M. Patel — *2023 - Present*

Indian Institute of Technology Jodhpur — Jodhpur, India
Bachelors in Computer Science and Engineering — *2019 - 2023*

Research Experience

Salesforce — Palo Alto, CA
Ph.D. Research Intern — *Summer 2026*
Advised by Becky Xiangyu Peng, Prafulla Kumar Choubey, Jiaxin Zhang, Silvio Savarese, and Chien-Sheng Wu
- Developed efficient multimodal agentic search methods and leveraged intermediate steps for calculating reward.
- Analyzed evaluation bias and explored Tree-RL for multimodal web-search.

Apple Inc. — Seattle, WA
Ph.D. Research Intern — *Spring 2026*
Advised by Tian Cao, Zheng Yang, Peng Zhou, Lunbo Xu, Xiaohui Tao, and Zhe Gan
- Developed a multi-image, vision-centric benchmark for evaluating multimodal agentic search.
- Analyzed failure modes of multimodal models in agentic search settings involving multiple visual entities.

Apple Inc. — Seattle, WA
Ph.D. Research Intern — *Summer 2025*
Advised by Yang Xu, Kavya Nerella, Navid Shiee, Peter Grasch, Chao Jia, Yinfei Yang, and Zhe Gan
- Equipped multimodal LLMs with web-search tools and enabled reasoning over iterative search calls.
- Developed methods to incentivize self-correction and self-verification following an initial model response.

Johns Hopkins University — Baltimore, MD
Research Assistant, Vision and Image Understanding Lab — *2023 – Present*
Advised by Dr. Vishal M. Patel
- Investigating fundamental challenges in text-to-video (T2V) models, including text-rich video generation.
- Developing multi-agent systems for automated and comprehensive facial analysis.

Indian Institute of Technology Jodhpur — Jodhpur, India
Student Researcher, IAB Lab — *2022 – 2023*
Advised by Prof. Richa Singh and Prof. Mayank Vatsa
- Studied deepfake video generation under challenging conditions, including multi-face and low-resolution videos.
- Introduced deepfake phylogeny by modeling the evolution of deepfakes through sequential face-swapping operations.
- Developed DeSI, a method for identifying deepfake videos on social media and analyzing their propagation.

Publications

1. **DeepMMSearch-R1: Empowering Multimodal LLMs in Multimodal Web Search**
   Kartik Narayan, *Yang Xu*, *Tian Cao*, *Kavya Nerella*, *Vishal M. Patel*, *Navid Shiee*, *Peter Grasch*, *Chao Jia*, *Yinfei Yang*, *Zhe Gan*
   Technical Report. [arXiv]
2. **MultiImage-MMSearch: Multi-Image Vision Centric Benchmark for Multimodal Agentic Search in MLLMs**
   Kartik Narayan, *Tian Cao*, *Zheng Yang*, *Peng Zhou*, *Vishal M. Patel*, *Lunbo Xu*, *Xiaohui Tao*, *Zhe Gan*
   Under Review.

3. **Lessons from Agentic Post-Training of MLLMs for Multimodal Web Search**
   Kartik Narayan, *Becky Xiangyu Peng*, *Prafulla Kumar Choubey*, *Jiaxin Zhang*, *Bin Lei*, *Vishal M. Patel*, *Silvio Savarese*, *Chien-Sheng Wu*
   Work in Progress.
4. **Fork Where the Model Changes Its Mind: Belief-Shift Branching for Tree-Structured Reinforcement Learning**
   Bin Lei, *Yu Li*, *Prafulla Kumar Choubey*, *Jiaxin Zhang*, *Becky Xiangyu Peng*, *Qinyuan Ye*, ***Kartik Narayan***, *Caiwen Ding*, *Silvio Savarese*, *Chien-Sheng Wu*
   Under Review.
5. **FaceMoE: Mixture of Experts for Low-resolution Face Recognition**
   Kartik Narayan, *Vishal M. Patel*
   European Conference on Computer Vision 2026 (ECCV 2026)
6. **Silhouette-based Gait Foundation Model**
   Dingqiang Ye, *Chao Fan*, ***Kartik Narayan***, *Bingzhe Wu*, *Chengwen Luo*, *Jianqiang Li*, *Vishal M. Patel*
   European Conference on Computer Vision 2026 (ECCV 2026 Spotlight)
7. **RestoreVAR: Visual Autoregressive Generation for All-in-One Image Restoration**
   Sudarshan Rajagopalan, ***Kartik Narayan***, *Vishal M. Patel*
   International Conference on Learning Representations 2026 (ICLR 2026)
8. **PromptPrint: Behavioral Biometrics Through Natural Language Prompting in LLMs**
   Shaiv Patel, ***Kartik Narayan***, *Vishal M. Patel*
   IEEE/IAPR International Joint Conference on Biometrics (IJCB 2026)
9. **TransFIRA: Transfer Learning for Face Image Recognizability Assessment**
   Allen Tu, ***Kartik Narayan***, *Joshua Gleason*, *Jennifer Xu*, *Matthew Meyn*, *Tom Goldstein*, *Vishal M. Patel*
   20th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2026)
10. **Training-Free Stylized Abstraction**
    *Aiman Rahman**, ***Kartik Narayan****, *Vishal M. Patel*
    Under Review. [arXiv]
11. **INFER: Implicit Neural Features for Exposing Realism**
    Dhananjaya Jayasundara, ***Kartik Narayan***, *Vishal M. Patel*
    Under Review. [paper]
12. **FaceXFormer: A Unified Transformer for Facial Analysis**
    Kartik Narayan*, *Vibashan VS**, *Rama Chellappa*, *Vishal M. Patel*
    Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV 2025)
13. **SegFace: Face Segmentation of Long-Tail Classes**
    Kartik Narayan, *Vibashan VS*, *Vishal M. Patel*
    Association for the Advancement of Artificial Intelligence (AAAI 2025)
14. **PETALface: Parameter Efficient Transfer Learning for Low-resolution Face Recognition**
    Kartik Narayan, *Nithin Gopalakrishnan Nair*, *Jennifer Xu*, *Rama Chellappa*, *Vishal M. Patel*
    IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025 Oral)
15. **Investigating Social Biases in Multimodal LLMs**
    *Malsha Perera**, ***Kartik Narayan****, *Vishal M. Patel*
    19th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2025)
16. **Improved Representation Learning for Unconstrained Face Recognition**
    *Nithin Gopalakrishnan Nair**, ***Kartik Narayan****, *Maitreya Suin*, *Ram Prabhakar*, *Jennifer Xu*, *Soraya Stevens*, *Joshua Gleason*, *Nathan Shindman*, *Rama Chellappa*, *Vishal M. Patel*
    19th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2025)
17. **Hyp-OC: Hyperbolic One Class Classification for Face Anti-Spoofing**
    Kartik Narayan, *Vishal M. Patel*
    18th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2024)
18. **DF-Platter: Multi-Face Heterogeneous Deepfake Dataset**
    Kartik Narayan, *Harsh Agarwal*, *Kartik Thakral*, *Surbhi Mittal*, *Mayank Vatsa*, *Richa Singh*
    Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2023)
19. **DeePhyNet: Towards Detecting Phylogeny in Deepfakes**
    Kartik Thakral, *Harsh Agarwal*, ***Kartik Narayan***, *Surbhi Mittal*, *Mayank Vatsa*, *Richa Singh*
    IEEE Transactions on Biometrics, Behavior, and Identity Science (T-BIOM)
20. **DeePhy: On Deepfake Phylogeny**
    Kartik Narayan, *Harsh Agarwal*, *Kartik Thakral*, *Surbhi Mittal*, *Mayank Vatsa*, *Richa Singh*

IEEE International Joint Conference on Biometrics (IJCB 2022)

21. **DeSI: Deepfake Source Identifier for Social Media**
Kartik Narayan, *Harsh Agarwal*, *Surbhi Mittal*, *Kartik Thakral*, *Suman Kundu*, *Mayank Vatsa*, *Richa Singh*
Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW 2022)
22. **Using Epidemic Modeling, Machine Learning & Control Feedback Strategy for Policy Management of COVID-19**
Kartik Narayan, *Heena Rathore*, *Faycal Znidi*
IEEE Access 10 (2022): 98244–98258
23. **Leveraging Ambient Sensing for the Estimation of Curiosity-Driven Human Crowd**
Anirban Das, ***Kartik Narayan***, *Suchetana Chakraborty*
IEEE International Systems Conference (SysCon 2022)

Academic Services

- **Invited Reviewer**: CVPR, ICCV, ECCV, AAAI, NeurIPS, ICLR, ICML, TPAMI, WaCV, IJCB, TIFS, TBIOM, IJCV, PR

Skills Summary

- **Programming**: Python, C/C++
- **ML/DL**: PyTorch, TensorFlow, Keras, OpenCV, Sklearn, Numpy, Pandas, Matplotlib
- **Development**: HTML, CSS, React.js, Node.js, Bootstrap, Firebase, MongoDB, ReactNative
- **Tools**: Docker, Kubernetes, GIT, AWS

Teaching Experience

Teaching Assistantship: Assisted in teaching by conducting weekly lab sessions, holding special doubt-clearing sessions, preparing quizzes, and grading assignments for the following courses:

- Deep Learning [*Spring 2023*]
- Introduction to Machine Learning [*Fall 2022*]
- Pattern Recognition and Machine Learning [*Spring 2022*]

Co-curricular Activities

- **Internship Head, CSE at the placement cell of IIT Jodhpur** [*2021 - 2022*]
Coordinated with companies to invite them for campus internships and placements.
- **Head of technical events in Prometeo, IIT Jodhpur** [*2021 - 2022*]
Led a team of 50+ students to organize technical competitions, attracting over 1,500 participants from across India.
- **Secretary of PHEME, The IITJ Newsletter Club** [*2020 - 2021*]
Managed a club of 80+ members responsible for creating newsletters, reports, conducting surveys, etc.
- **Student Guide at Student Wellbeing Committee, IITJ** [*2020 - 2023*]
Mentored 10 freshmen from diverse backgrounds to support their transition into college, guiding them in both academic and non-academic pursuits.